

Report
Of
Software Engineering Job Simulation

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, HARSHIT BHARDWAJ do hereby declare that the internship report titled “**Software Engineering Job Simulation** ” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: HARSHIT BHARDWAJ

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Forage and Tom Brunskill, CEO and Co-Founder of Forage, for providing the platform to complete this Software Engineering Job Simulation.

I express my gratitude to my parents for their blessings.

Signature

Student Name :HARSHIT BHARDWAJ

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3. INTERNSHIP COMPLETION CERTIFICATE

JPMORGAN CHASE & CO.



Inspiring and empowering
future professionals

Harshit Bhardwaj

Software Engineering Job Simulation

Certificate of Completion
August 4th, 2026

Over the period of July 2026 to August 2026, Harshit Bhardwaj has completed practical tasks in:

Project Setup
Kafka Integration
H2 Integration
REST API Integration
REST API Controller

A handwritten signature in black ink, appearing to read "Tom Brunskill".

Tom Brunskill
CEO, Co-Founder of
Forage

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Software Engineering Job Simulation provided by Forage. The virtual internship was carried out over the period of July 2026 to August 2026. The primary aim of this program was to provide hands-on experience with enterprise-level software engineering practices, specifically focusing on backend system architecture, data streaming, and API development. By completing tasks in a simulated corporate environment, the project bridged the gap between theoretical computer science concepts and practical, industry-standard implementation.

4.2 Organization Profile

The simulation was hosted by Forage, an organization dedicated to inspiring and empowering future professionals by offering virtual job simulations in collaboration with top global companies. The tasks were modeled after real-world engineering challenges faced by developers building robust, scalable financial technology solutions.

4.3 Problem Statement

Modern enterprise applications require highly scalable, decoupled, and robust backend systems to process large volumes of data in real-time. The core challenge addressed in this simulation was to construct a foundational backend

architecture capable of handling asynchronous messaging and lightweight data storage, while exposing these capabilities through a standardized web interface.

4.4 Project Objectives

The core objectives of this job simulation were to successfully complete practical tasks across five key technical domains:

- **Project Setup:** To establish the initial development environment, configure necessary dependencies, and initialize the foundational codebase for a scalable backend application.
- **Kafka Integration:** To implement a robust message-broker system using Apache Kafka, enabling asynchronous data streaming and decoupled communication between different system components.
- **H2 Integration:** To configure and utilize an H2 in-memory database, allowing for rapid testing, seamless application data storage, and efficient query execution during the development phase.
- **REST API Integration:** To connect various internal backend services and ensure they communicate effectively using RESTful architectural principles.
- **REST API Controller:** To design and build the controller layer that handles incoming HTTP client requests, orchestrates backend logic, and returns appropriate HTTP responses.

4.5 Scope of the Project

The scope of this project was entirely focused on backend software development, system integration, and data flow management. It encompassed the complete lifecycle of setting up a robust data pipeline, starting from transient database configuration via H2 and event streaming setup via Kafka.

It culminated in the exposure of these internal services through a fully functional REST API Controller layer. Front-end development, user interface (UI) design, user experience (UX) optimization, and cloud deployment pipelines (CI/CD) were deliberately kept outside the scope of this specific simulation to maintain a strict focus on core backend engineering.

4.6 Technologies and Tools Used

Based on the practical tasks completed during the simulation, the following technologies and architectural concepts were central to the project:

- **Messaging and Event Streaming:** Kafka integration was utilized to establish an event-driven architecture, utilizing producers and consumers to handle data asynchronously.
- **Database Management:** H2 Integration was used to deploy an open-source, lightweight relational database management system written in Java, designed specifically for fast, in-memory data persistence during runtime.
- **Web Architecture:** RESTful API principles were strictly followed to ensure stateless, reliable client-server communication using standard HTTP methods (GET, POST, PUT, DELETE).
- **Platform:** The Forage virtual simulation environment provided the infrastructure and task parameters for the project.

4.7 System Architecture

The architecture implemented during the simulation followed a standard, highly efficient multi-tiered backend model designed for separation of concerns:

- **Client Interface / Controller Layer:** Handled entirely by the REST API Controller, this layer acted as the sole entry point for all HTTP requests, ensuring proper routing and payload validation.
- **Integration & Business Logic Layer:** Facilitated by the REST

API Integration, this intermediary layer processed validated requests, applied necessary business rules, and orchestrated tasks between the messaging and data layers.

- **Event Streaming Layer:** Powered by Kafka Integration, this layer managed high-throughput message queues, allowing the system to process background tasks and data streams without blocking the main application execution thread.
- **Data Access Layer:** Powered by H2 Integration, this layer provided a fast, transient storage solution for the application's relational data, ensuring quick read and write operations during the simulation.

4.8 Methodology

The project followed a sequential, iterative methodology over the course of December 2025 to January 2026, mimicking an agile sprint environment.

1. **Phase 1 (Initialization):** The project began with the foundational Project Setup to ensure all build tools and dependencies were correctly mapped and functional.
2. **Phase 2 (Infrastructure Setup):** The core data and messaging backbones were then established through dedicated H2 Integration for storage and Kafka Integration for event handling.
3. **Phase 3 (API Development):** Finally, the disparate systems were tied together and made securely accessible to external clients via comprehensive REST API Integration and the programmatic development of the REST API Controller endpoints.

4.9 Expected Outcomes

By the end of the job simulation, several highly technical and practical outcomes were successfully achieved:

- Mastery of initial project configuration and dependency management for enterprise-scale backend development.

- A fully working implementation of Kafka architecture, demonstrating the ability to securely produce and consume data streams.
- A successfully integrated H2 database environment capable of performing efficient CRUD (Create, Read, Update, Delete) operations in memory.
- A robust, functional REST API Controller capable of processing web requests, managing state, and interacting seamlessly with both the underlying messaging queues and database services.

4.10 Certificates of Completion and Communication Proof

JPMORGAN CHASE & CO.



Harshit Bhardwaj

Software Engineering Job Simulation

Certificate of Completion
August 4th, 2026

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